

Cisco Webex Integration Registration & Consent Record

Document Reference: DOC-06

System: Case Ace v2.0 Client Consultation Recording & Note Drafting System

Organisation: Citizens Advice Wandsworth (CAW)

Integration Type: Enterprise Webex Calling Ingest & Audio Processing

Status: Active / Production Pilot Registration

Classification: Official-Sensitive / Governance Pack

1. Application Registration & Governance Details

2. Granted OAuth Scopes & Technical Justification

3. Webhook Security & Ingestion Architecture

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sequenceDiagram
    autonumber
    participant Client as Client Adviser Browser
    participant Webex as Cisco Webex Calling
    participant Backend as CAW Node Backend
    participant KMS as Webex KMS Service

    Webex->>Backend: Webhook Event: telephony_call_started (Signed with HMAC-SHA256)
    Backend->>Backend: Verify X-Spark-Signature using Shared Webhook Secret
    Backend->>Client: Push Call Notification to Authenticated Adviser (WebSocket)
    Client->>KMS: Request Ephemeral SRTP Media Decryption Key (OAuth spark:kms)
    KMS-->>Client: Return Ephemeral Key (Volatile RAM)
    Webex-->>Client: Stream Encrypted SRTP Audio Packets (TLS 1.3)
    Client->>Client: Decrypt SRTP to PCM in Volatile RAM Buffer (Zero Disk Writes)
    Note over Client: In-Browser Whisper WASM Pass 1 & Redaction Gate
    Webex->>Backend: Webhook Event: telephony_call_ended
    Client->>Client: Close Stream & Flush/Zero SRTP Decryption Keys
```

4. Operational Invariants & Compliance Proofs

Cloud Recording Prohibited:

Webex Organisation administrative policy enforces `auto_record`: disabled and `allow_cloud_recording`: false. No audio is stored in Cisco Webex Cloud Storage.

Local Media Decryption:

Media packets are streamed directly from Webex telephony nodes to the adviser's local browser session over TLS 1.3 / SRTP. No raw telephony audio is intercepted or stored on the CAW backend server.

Webhook Cryptographic Integrity:

Every inbound webhook notification is authenticated via HMAC-SHA256 header validation (`backend/src/routes/webex.ts`). Payloads lacking valid signatures are immediately rejected with HTTP 401.

Immediate Stream Destruction:

When the telephony call terminates or the client hangs up, the media stream is closed and the SRTP decryption keys are immediately zeroed from browser memory.